A Additional Depth Results

Table 1: OccuFly test MDE comparison grouped by altitude, with Metric3Dv2-OccuFly and DepthAnything2-OccuFly altitude-wise values appended from OccuFly Table 12 [1]. Light pink marks the best RMSE within each altitude group.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
30	Linear Probe RGB	0.743	0.952	0.986	0.205	6.280	5.391	0.176
	Linear Probe Video 4 frame clip	0.512	0.933	0.979	0.266	7.942	7.002	0.177
	SFP_FiLM_DPT	0.889	0.961	0.991	0.111	4.022	2.847	0.166
	DPT	0.415	0.889	0.940	0.295	9.505	8.238	0.350
	FiLM_DPT_RGB	0.917	0.980	0.999	0.097	3.448	2.571	0.132
	DTA_FiLM_DPT_Video	0.944	0.982	0.999	0.081	2.939	2.051	0.124
	Metric3Dv2-OccuFly [1]	0.460	0.757	0.992	0.347	10.917	10.215	0.101
	DepthAnything2-OccuFly [1]	0.919	0.982	0.998	0.103	3.108	2.666	0.086
40	Linear Probe RGB	0.730	0.943	0.998	0.180	5.851	4.804	0.217
	Linear Probe Video 4 frame clip	0.720	0.924	0.996	0.185	5.763	4.510	0.207
	SFP_FiLM_DPT	0.729	0.911	0.988	0.186	5.665	4.056	0.220
	DPT	0.577	0.794	0.905	0.239	9.106	6.916	0.391
	FiLM_DPT_RGB	0.761	0.945	0.995	0.158	4.804	3.588	0.186
	DTA_FiLM_DPT_Video	0.737	0.938	0.997	0.173	5.191	3.800	0.195
	Metric3Dv2-OccuFly [1]	0.209	0.802	0.981	0.395	13.138	12.625	0.096
	DepthAnything2-OccuFly [1]	0.795	0.957	0.998	0.148	4.486	3.823	0.119
50	Linear Probe RGB	0.524	0.978	1.000	0.192	9.981	8.728	0.174
	Linear Probe Video 4 frame clip	0.741	0.994	1.000	0.161	8.244	7.158	0.167
	SFP_FiLM_DPT	0.858	0.995	1.000	0.110	5.038	3.777	0.138
	DPT	0.375	0.754	0.872	0.285	14.243	12.179	0.378
	FiLM_DPT_RGB	0.873	0.995	1.000	0.113	4.872	3.879	0.129
	DTA_FiLM_DPT_Video	0.817	0.998	1.000	0.122	5.213	4.165	0.135
	Metric3Dv2-OccuFly [1]	0.283	0.831	0.987	0.379	15.373	14.950	0.099
	DepthAnything2-OccuFly [1]	0.844	0.997	1.000	0.129	5.985	5.261	0.114

Table 2: OccuFly test MDE comparison grouped by scene. Light pink marks the best RMSE within each scene group.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
scene_08	Linear Probe RGB	0.752	0.985	0.999	0.159	6.972	5.715	0.187
	Linear Probe Video 4 frame clip	0.665	0.983	0.999	0.183	7.465	6.358	0.192
	SFP_FiLM_DPT	0.976	0.998	1.000	0.069	3.014	2.468	0.078
	DPT	0.601	0.946	0.976	0.212	8.949	7.353	0.271
	FiLM_DPT_RGB	0.975	0.997	1.000	0.067	2.934	2.392	0.079
	DTA_FiLM_DPT_Video	0.982	0.998	1.000	0.055	2.566	2.010	0.071
scene_09	Linear Probe RGB	0.610	0.947	0.995	0.201	8.121	6.757	0.236
	Linear Probe Video 4 frame clip	0.702	0.941	0.993	0.192	7.129	5.851	0.229
	SFP_FiLM_DPT	0.741	0.935	0.990	0.173	5.805	4.242	0.206
	DPT	0.415	0.734	0.866	0.289	12.343	9.983	0.430
	FiLM_DPT_RGB	0.776	0.960	0.997	0.155	5.147	3.979	0.174
	DTA_FiLM_DPT_Video	0.734	0.958	0.998	0.169	5.542	4.287	0.179

B Detailed SSC Results

Additional class-wise and altitude-wise SSC results are provided here to keep the main paper focused on the principal trends.

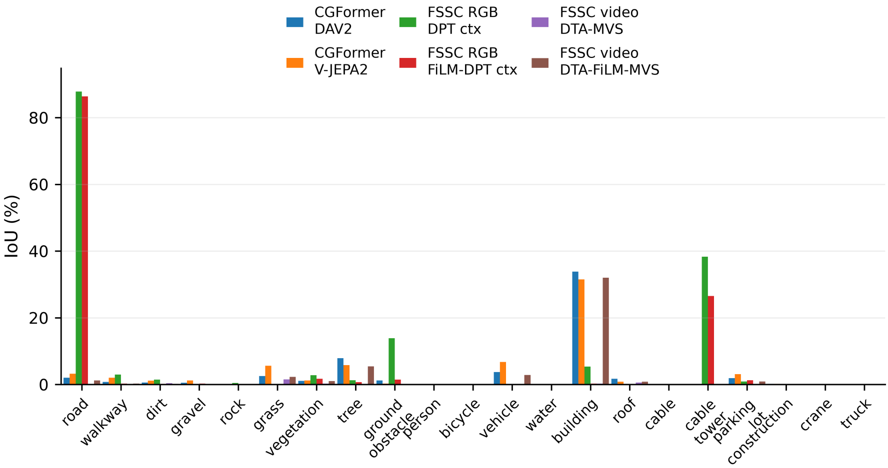

Fig. 1: OccuFly SSC per-class IoU bar chart for the evaluated CGFormer and FoundationSSC variants.

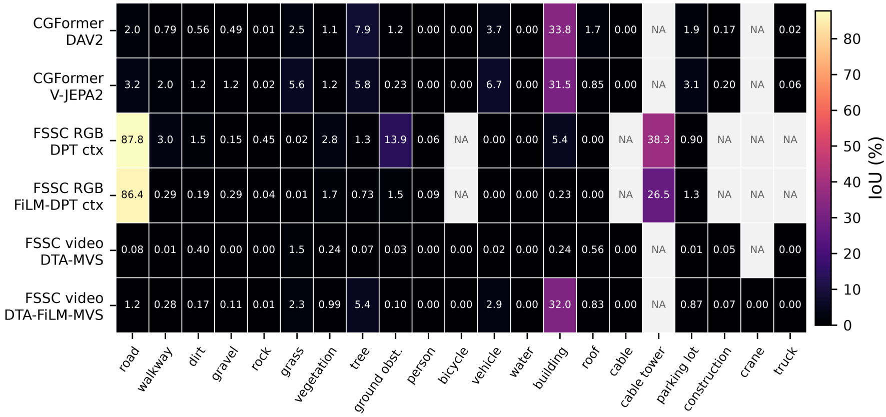

Fig. 2: OccuFly SSC per-class IoU heatmap for the evaluated CGFormer and FoundationSSC variants.

Table 3: OccuFly SSC benchmark comparison by altitude. Published Symphonies and DISC values follow OccuFly Table 4 [1]; CGFormer and FoundationSSC values are from the local SSC benchmark report. For our variants, light beige marks the best SC IoU and light pink marks the best SSC mIoU within each altitude group and the overall All group.

Altitude	Method	SC IoU ↑	SSC mIoU ↑
Published OccuFly Table 4 baselines			
50 m	Symphonies [2]	15.88	0.58
50 m	DISC [3]	31.10	2.20
40 m	Symphonies [2]	10.71	0.52
40 m	DISC [3]	27.85	1.77
30 m	Symphonies [2]	13.22	0.76
30 m	DISC [3]	26.88	2.23
All	Symphonies [2]	13.68	0.58
All	DISC [3]	29.52	2.04
Our CGFormer and FoundationSSC variants			
30 m	CGFormer DAV2 prior	36.10	4.22
30 m	CGFormer V-JEPA 2.1 FiLM__DPT__RGB prior	36.70	4.39
30 m	FSSC RGB ctx DPT, depth FiLM-DPT	33.21	5.16
30 m	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	6.74	0.26
30 m	FSSC video ctx DTA-DPT, depth DTA-MVS	2.84	0.18
30 m	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	35.59	5.06
40 m	CGFormer DAV2 prior	38.42	2.86
40 m	CGFormer V-JEPA 2.1 FiLM__DPT__RGB prior	29.05	2.12
40 m	FSSC RGB ctx DPT, depth FiLM-DPT	27.85	4.08
40 m	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	22.15	1.69
40 m	FSSC video ctx DTA-DPT, depth DTA-MVS	12.30	0.87
40 m	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	29.22	3.97
50 m	CGFormer DAV2 prior	35.56	2.81
50 m	CGFormer V-JEPA 2.1 FiLM__DPT__RGB prior	42.78	4.06
50 m	FSSC RGB ctx DPT, depth FiLM-DPT	21.16	2.53
50 m	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	19.51	0.75
50 m	FSSC video ctx DTA-DPT, depth DTA-MVS	5.21	0.21
50 m	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	22.28	2.59
All	CGFormer DAV2 prior	36.77	3.05
All	CGFormer V-JEPA 2.1 FiLM__DPT__RGB prior	36.51	3.31
All	FSSC RGB ctx DPT, depth FiLM-DPT	37.68	3.46
All	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	35.47	1.73
All	FSSC video ctx DTA-DPT, depth DTA-MVS	24.55	0.17
All	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	32.55	2.36

References

1. Gross, M., Matha, S.B., Fahmy, A., Song, R., Cremers, D., Meess, H.: Occufly: A 3d vision benchmark for semantic scene completion from the aerial perspective. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2026) 1, 4
2. Jiang, H., Cheng, T., Gao, N., Zhang, H., Lin, T., Liu, W., Wang, X.: Symphonize 3d semantic scene completion with contextual instance queries. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024) 4
3. Liu, E., Yu, E., Chen, S., Tao, W.: Disentangling instance and scene contexts for 3d semantic scene completion (2025) 4